

Multiple Choice

1. **Answer:** A

Options: A) 0.87; B) -0.23; C) 0.07; D) 0.57; E) 0.97

Note: Correct option: A - 0.87

2. **Answer:** D

Options: A) is amplified; B) converts to magnetic energy; C) is parallel to the electric field; D) changes; E) is at right angles to the electric field

Note: Correct option: D - changes

3. **Answer:** A

Options: A) 14.1 eV; B) 10.6 eV; C) 11.2 eV; D) 17.0 eV; E) 15.3 eV

Note: Correct option: A - 14.1 eV

4. **Answer:** A

Options: A) ${}^8_4\text{Be}$; B) ${}^{13}_6\text{C}$; C) ${}^7_4\text{Be}$; D) ${}^6_3\text{Li}$; E) ${}^5_2\text{He}$

Note: Correct option: A - ${}^8_4\text{Be}$

4. **Answer:** B

Options: A) an average of 10 times, with an rms deviation of about 5; B) an average of 10 times, with an rms deviation of about 3; C) an average of 10 times, with an rms deviation of about 0.1; D) an average of 10 times, with an rms deviation of about 1; E) an average of 10 times, with an rms deviation of about 6

Note: Correct option: B - an average of 10 times, with an rms deviation of about 3

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'proportional to amplitude'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Current and power'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 2 C. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.068 K. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key